Quaternion Economic Field Theory

A Unified Framework for
Two Interacting Complex Economies

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops *Quaternion Economic Field Theory* as a two-economy generalization of Complex Economic Field Theory. Whereas a complex economic field represents one economy through a real material component and one latent imaginary component, a quaternionic economic field represents a two-economy system through four coupled coordinates: a shared material baseline, the latent field of economy A, the latent field of economy B, and the bilateral interaction field AB. The basic object is

$$q = q_0 + \mathbf{i}q_A + \mathbf{j}q_B + \mathbf{k}q_{AB} \in \mathbb{H},$$

where $\mathbb{H}$ denotes the algebra of quaternions.

The model uses quaternionic norms to measure effective capacity, unit quaternions to measure orientation, and quaternionic coherence to measure alignment between domestic, bilateral, monetary, geopolitical, and military fields. Because quaternion multiplication is non-commutative, the framework naturally represents path dependence: policy sequences, sanctions, military escalation, monetary interventions, and supply-chain adjustments need not commute. The paper develops axioms, production theory, bilateral interference theorems, macro-financial equations, exchange-rate dynamics, trade and supply-chain transmission models, geopolitical and warfare-economic applications, statistical-mechanical order parameters, econometric identification strategies, and AI/ML algorithms. The central claim is that a two-economy system is not merely $A + B$, but $A + B + AB$: the bilateral field is itself an irreducible source of power, instability, growth, and conflict.

The paper ends with “The End”

Keywords: quaternion economics, complex economic fields, two-economy model, bilateral interaction, non-commutative policy, geopolitics, warfare economics, macro-finance, artificial intelligence, statistical mechanics.

Contents

List of Figures

List of Tables

1 Introduction

Complex Economic Field Theory represents a single economy as a coupled material-latent system, where a complex field

$$z = R + iI$$

contains an observable material component R and a latent informational, institutional, cognitive, or strategic component I [?]. The modulus $|z|$ measures effective capacity, while the phase $\arg(z)$ measures alignment, coherence, expectation, legitimacy, or strategic orientation.

A two-economy world requires a richer algebra. If economy A and economy B each possess internal latent fields, and if their bilateral interaction is not reducible to either domestic economy alone, then a natural representation is quaternionic:

$$q = q_0 + \mathbf{i}q_A + \mathbf{j}q_B + \mathbf{k}q_{AB}.$$

Here q_0 is the material or common baseline, $\mathbf{i}q_A$ is the latent field internal to economy A, $\mathbf{j}q_B$ is the latent field internal to economy B, and $\mathbf{k}q_{AB}$ is the bilateral interaction field.

The economic importance of the $\mathbf{k}$-coordinate is substantial. It can represent trade dependence, supply-chain exposure, currency linkage, sanctions, capital-flow stress, migration, alliance complementarity, cyber interaction, information warfare, or bilateral hostility. The two-economy system is therefore not merely an additive pair. It is a structured object:

$$A + B + AB.$$

Quaternion algebra also introduces non-commutativity:

$$pq \neq qp$$

in general. This is not a mathematical inconvenience; it is a political-economic advantage. Many economic and strategic processes are order-dependent. Monetary easing followed by capital controls differs from capital controls followed by monetary easing. Sanctions followed by supply-chain rerouting differ from supply-chain rerouting followed by sanctions. Military mobilization followed by diplomacy differs from diplomacy followed by military mobilization. Quaternion Economic Field Theory provides an algebraic language for this path dependence.

The paper makes seven contributions. First, it develops a quaternionic two-economy state space. Second, it defines quaternionic capacity, orientation, coherence, and decoherence. Third, it constructs a phase-coherent production model for two interacting economies. Fourth, it proves bilateral interference and coherence-production theorems. Fifth, it applies the framework to macroeconomics, finance, exchange rates, trade, geopolitics, international relations, warfare, and sanctions. Sixth, it develops statistical-mechanical and econometric tools for identifying quaternionic fields. Seventh, it presents AI/ML algorithms for forecasting, crisis detection, and policy learning.

2 Quaternion Algebra

2.1 Definition

The quaternion algebra is

$$\mathbb{H} = \{q_0 + \mathbf{i}q_1 + \mathbf{j}q_2 + \mathbf{k}q_3 : q_0, q_1, q_2, q_3 \in \mathbb{R}\}.$$

The basis elements satisfy

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1,$$

$$\mathbf{ij} = \mathbf{k}, \qquad \mathbf{jk} = \mathbf{i}, \qquad \mathbf{ki} = \mathbf{j},$$

and

$$\mathbf{ji} = -\mathbf{k}, \qquad \mathbf{kj} = -\mathbf{i}, \qquad \mathbf{ik} = -\mathbf{j}.$$

$\cdot$	1	$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$
1	1	$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$
$\mathbf{i}$	$\mathbf{i}$	-1	$\mathbf{k}$	$-\mathbf{j}$
$\mathbf{j}$	$\mathbf{j}$	$-\mathbf{k}$	-1	$\mathbf{i}$
$\mathbf{k}$	$\mathbf{k}$	$\mathbf{j}$	$-\mathbf{i}$	-1

Table 1: Quaternion multiplication table.

2.2 Conjugate, norm, and inverse

For

$$q = q_0 + \mathbf{i}q_1 + \mathbf{j}q_2 + \mathbf{k}q_3,$$

define the conjugate

$$\overline{q} = q_0 - \mathbf{i}q_1 - \mathbf{j}q_2 - \mathbf{k}q_3.$$

The norm is

$$|q| = \sqrt{q\overline{q}} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}.$$

If $q \neq 0$, its inverse is

$$q^{-1} = \frac{\overline{q}}{|q|^2}.$$

Proposition 1 (Norm multiplicativity)**.** *For all* $p, q \in \mathbb{H}$*,*

$$|pq| = |p||q|.$$

Proof. Using quaternion conjugation,

$$|pq|^2 = (pq)\overline{(pq)}.$$

Since

$$\overline{(pq)} = \overline{q}\,\overline{p},$$

we obtain

$$|pq|^2 = pq\overline{q}\,\overline{p} = p|q|^2\overline{p} = |q|^2 p\overline{p} = |p|^2|q|^2.$$

Taking square roots gives

$$|pq| = |p||q|.$$

□

2.3 Unit quaternions and orientation

If $q \neq 0$, define its unit orientation:

$$u_q = \frac{q}{|q|}.$$

Then

$$|u_q| = 1.$$

The set of unit quaternions is the three-sphere:

$$S^3 = \{u \in \mathbb{H} : |u| = 1\}.$$

Thus the complex model's phase circle S^1 is generalized to a quaternionic orientation sphere S^3.

3 Economic Interpretation of Quaternion Coordinates

For any strategic variable X, define the two-economy quaternionic field

$$\mathcal{X} = X_0 + \mathbf{i}X_A + \mathbf{j}X_B + \mathbf{k}X_{AB}.$$

The four coordinates have the following interpretation:

$$\begin{aligned} X_0 &= \text{common material or measured baseline}, \\ X_A &= \text{latent internal component of economy } A, \\ X_B &= \text{latent internal component of economy } B, \\ X_{AB} &= \text{bilateral interaction component.} \end{aligned}$$

Field	q_0	$\mathbf{i}q_A$	$\mathbf{j}q_B$	$\mathbf{k}q_{AB}$
$\mathcal{K}$	capital stock	A-technology	B-technology	supply-chain capital
$\mathcal{L}$	labor hours	A-skills	B-skills	migration linkage
$\mathcal{M}$	liquidity	A-confidence	B-confidence	exchange/debt stress
$\mathcal{P}$	state capacity	A-legitimacy	B-legitimacy	diplomatic relation
$\mathcal{G}$	material power	A-strategy	B-strategy	alliance/rivalry
$\mathcal{W}$	hardware	A-doctrine	B-doctrine	cyber/info conflict

Table 2: Economic interpretation of quaternion coordinates.

4 Axioms of Quaternion Economic Field Theory

Axiom 1 (Quaternionic two-economy decomposition)**.** *Every strategic two-economy variable is represented as*

$$\mathcal{X} = X_0 + \mathbf{i}X_A + \mathbf{j}X_B + \mathbf{k}X_{AB},$$

where X_0 is the shared material baseline, X_A is economy A's latent internal component, X_B is economy B's latent internal component, and X_{AB} is the bilateral interaction component.

Axiom 2 (Capacity principle)**.** *The effective capacity of a quaternionic field $\mathcal{X}$ is measured by*

$$|\mathcal{X}| = \sqrt{X_0^2 + X_A^2 + X_B^2 + X_{AB}^2}.$$

Axiom 3 (Orientation principle)**.** *The unit quaternion*

$$u_{\mathcal{X}} = \frac{\mathcal{X}}{|\mathcal{X}|}$$

measures the field's orientation in the four-dimensional material-latent-bilateral space.

Axiom 4 (Bilateral irreducibility)**.** *The bilateral component X_{AB} is not generally reducible to X_A or X_B. A two-economy system contains an interaction field that may independently affect output, inflation, trade, exchange rates, alliances, crisis, and conflict.*

Axiom 5 (Non-commutative policy principle)**.** *Policy and strategic operations are represented by quaternionic transformations. Since quaternion multiplication is generally non-commutative,*

$$UV \neq VU,$$

the order of policy actions may affect final economic and geopolitical outcomes.

Axiom 6 (Realization principle)**.** *Observed economic, political, financial, and military outcomes are real-valued. They are generated by realization maps*

$$\mathcal{R} : \mathbb{H}^n \to \mathbb{R}^m.$$

5 Quaternionic Coherence and Decoherence

Definition 1 (Quaternionic coherence). *For nonzero $p, q \in \mathbb{H}$, define coherence by*

$$C(p,q) = \frac{\mathrm{Re}(p\overline{q})}{|p||q|}.$$

Since

$$\mathrm{Re}(p\overline{q}) = p_0q_0 + p_1q_1 + p_2q_2 + p_3q_3,$$

the coherence $C(p,q)$ is the cosine similarity between the two quaternionic coordinate vectors.

Proposition 2 (Coherence bounds). *For all nonzero $p, q \in \mathbb{H}$,*

$$-1 \leq C(p,q) \leq 1.$$

Proof. Identifying p and q with vectors in $\mathbb{R}^4$,

$$\mathrm{Re}(p\overline{q}) = p \cdot q.$$

By the Cauchy-Schwarz inequality,

$$|p \cdot q| \leq |p||q|.$$

Dividing by $|p||q| > 0$ gives

$$-1 \leq C(p,q) \leq 1.$$

□

Definition 2 (Quaternionic decoherence). *For nonzero $p, q \in \mathbb{H}$, define decoherence distance by*

$$D(p,q) = 1 - C(p,q).$$

Thus

$$0 \leq D(p,q) \leq 2.$$

Perfect alignment corresponds to

$$C(p,q) = 1, \qquad D(p,q) = 0.$$

Orthogonality corresponds to

$$C(p,q) = 0, \qquad D(p,q) = 1.$$

Opposition corresponds to

$$C(p,q) = -1, \qquad D(p,q) = 2.$$

6 The Two-Economy Quaternionic State

Let the two-economy state be

$$\Psi_t = \begin{bmatrix} \mathcal{K}_{A,t} \\ \mathcal{L}_{A,t} \\ \mathcal{M}_{A,t} \\ \mathcal{P}_{A,t} \\ \mathcal{G}_{A,t} \\ \mathcal{W}_{A,t} \\ \mathcal{K}_{B,t} \\ \mathcal{L}_{B,t} \\ \mathcal{M}_{B,t} \\ \mathcal{P}_{B,t} \\ \mathcal{G}_{B,t} \\ \mathcal{W}_{B,t} \\ \mathcal{T}_{AB,t} \end{bmatrix} \in \mathbb{H}^{13}.$$

Here $\mathcal{T}_{AB,t}$ is the bilateral transmission field, including trade, finance, exchange-rate exposure, migration, technology transfer, alliance channels, sanctions, cyber interaction, and supply-chain dependence.

7 Quaternionic Production Theory

7.1 Realized output

For economy A, define realized output by

$$Y_{A,t} = |\mathcal{A}_{A,t}||\mathcal{K}_{A,t}|^{\alpha_A}|\mathcal{L}_{A,t}|^{\beta_A}|\mathcal{T}_{AB,t}|^{\tau_A}\Gamma_{A,t},$$

where

$$\alpha_A, \beta_A, \tau_A \geq 0.$$

Similarly,

$$Y_{B,t} = |\mathcal{A}_{B,t}||\mathcal{K}_{B,t}|^{\alpha_B}|\mathcal{L}_{B,t}|^{\beta_B}|\mathcal{T}_{BA,t}|^{\tau_B}\Gamma_{B,t}.$$

The coherence factor for economy A is

$$\Gamma_{A,t} = \exp\left[-\lambda_{KL,A}D(\mathcal{K}_{A,t}, \mathcal{L}_{A,t}) - \lambda_{MP,A}D(\mathcal{M}_{A,t}, \mathcal{P}_{A,t}) - \lambda_{AB,A}D(\mathcal{P}_{A,t}, \mathcal{P}_{B,t})\right],$$

with all λ's nonnegative. The corresponding coherence factor for economy B is

$$\Gamma_{B,t} = \exp\left[-\lambda_{KL,B}D(\mathcal{K}_{B,t}, \mathcal{L}_{B,t}) - \lambda_{MP,B}D(\mathcal{M}_{B,t}, \mathcal{P}_{B,t}) - \lambda_{BA,B}D(\mathcal{P}_{B,t}, \mathcal{P}_{A,t})\right].$$

Theorem 1 (Quaternionic coherence-production theorem)**.** *Let*

$$Y_A = |\mathcal{A}_A||\mathcal{K}_A|^{\alpha_A}|\mathcal{L}_A|^{\beta_A}|\mathcal{T}_{AB}|^{\tau_A} \exp\left[-\sum_{m=1}^{M} \lambda_m D(p_m, q_m)\right],$$

where $\lambda_m \geq 0$. Holding amplitudes fixed, Y_A is maximized with respect to orientations when

$$C(p_m, q_m) = 1$$

for every relevant pair (p_m, q_m) with $\lambda_m > 0$.

Proof. Since

$$D(p_m, q_m) = 1 - C(p_m, q_m),$$

and since $C(p_m, q_m) \leq 1$, each decoherence term satisfies

$$D(p_m, q_m) \geq 0.$$

The exponential factor is maximized when the exponent is maximized. Since the exponent is

$$-\sum_{m=1}^{M} \lambda_m D(p_m, q_m),$$

and $\lambda_m \geq 0$, it is maximized when every weighted decoherence term is minimized. Hence

$$D(p_m, q_m) = 0$$

for every $\lambda_m > 0$, equivalently

$$C(p_m, q_m) = 1.$$

□

7.2 Economic interpretation

Output is high when three conditions hold:

large domestic factor amplitudes,

large bilateral transmission amplitude,

and

high quaternionic coherence.

Thus a large capital stock may underperform if it is misaligned with labor, money, institutions, or the bilateral field. Conversely, moderate material capacity may produce high realized output when orientations are coherent.

8 Bilateral Interference

Theorem 2 (Bilateral interference theorem)**.** *Let $Q_A, Q_B \in \mathbb{H}$ be two economy-level fields. Define*

$$Q_{AB} = Q_A + Q_B.$$

Then

$$|Q_{AB}|^2 = |Q_A|^2 + |Q_B|^2 + 2|Q_A||Q_B|C(Q_A, Q_B).$$

Proof. We have

$$|Q_A + Q_B|^2 = (Q_A + Q_B)(\overline{Q_A} + \overline{Q_B}).$$

Taking the real scalar value of the norm,

$$|Q_A + Q_B|^2 = |Q_A|^2 + |Q_B|^2 + \mathrm{Re}(Q_A\overline{Q_B}) + \mathrm{Re}(Q_B\overline{Q_A}).$$

Since

$$\mathrm{Re}(Q_A\overline{Q_B}) = \mathrm{Re}(Q_B\overline{Q_A}),$$

we obtain

$$|Q_A + Q_B|^2 = |Q_A|^2 + |Q_B|^2 + 2\,\mathrm{Re}(Q_A\overline{Q_B}).$$

Using the definition of coherence,

$$\mathrm{Re}(Q_A\overline{Q_B}) = |Q_A||Q_B|C(Q_A, Q_B).$$

Therefore

$$|Q_{AB}|^2 = |Q_A|^2 + |Q_B|^2 + 2|Q_A||Q_B|C(Q_A, Q_B).$$

□

Corollary 1 (Constructive and destructive bilateral fields)**.** *If $C(Q_A, Q_B) > 0$, the bilateral combination is constructive. If $C(Q_A, Q_B) = 0$, it is orthogonal. If $C(Q_A, Q_B) < 0$, it is destructive.*

9 Quaternionic Dynamics

Let a quaternionic field q_t evolve according to

$$\dot{q}_t = g_t q_t + \Omega_{L,t} q_t + q_t \Omega_{R,t} + \eta_t,$$

where $g_t \in \mathbb{R}$ controls amplitude growth, while

$$\Omega_{L,t} = \omega_{L,i,t}\mathbf{i} + \omega_{L,j,t}\mathbf{j} + \omega_{L,k,t}\mathbf{k},$$

and

$$\Omega_{R,t} = \omega_{R,i,t}\mathbf{i} + \omega_{R,j,t}\mathbf{j} + \omega_{R,k,t}\mathbf{k}$$

are pure imaginary quaternions controlling left and right orientation rotations.

9.1 Quaternion exponential

For a pure imaginary quaternion

$$\Omega = \omega_i \mathbf{i} + \omega_j \mathbf{j} + \omega_k \mathbf{k},$$

let

$$|\Omega| = \sqrt{\omega_i^2 + \omega_j^2 + \omega_k^2}.$$

If $\Omega \neq 0$, then

$$e^{\Omega t} = \cos(|\Omega| t) + \frac{\Omega}{|\Omega|} \sin(|\Omega| t).$$

Proposition 3 (Pure orientation rotation)**.** *If*

$$\dot{q}_t = q_t \Omega,$$

where Ω is pure imaginary and constant, then

$$q_t = q_0 e^{\Omega t}.$$

Moreover,

$$|q_t| = |q_0|.$$

Proof. The differential equation has solution

$$q_t = q_0 e^{\Omega t}.$$

By norm multiplicativity,

$$|q_t| = |q_0||e^{\Omega t}|.$$

For pure imaginary Ω,

$$|e^{\Omega t}| = 1.$$

Thus

$$|q_t| = |q_0|.$$

□

10 Non-Commutative Policy

Let policies be represented by unit quaternions $U, V \in \mathbb{H}$. Applying U followed by V gives

$$q' = VUq.$$

Applying V followed by U gives

$$q'' = UVq.$$

Since generally

$$VU \neq UV,$$

the final economic state depends on the order of policy.

Theorem 3 (Policy-sequence non-commutativity)**.** *There exist policy transformations $U, V \in \mathbb{H}$ and states $q \in \mathbb{H}$ such that*

$$VUq \neq UVq.$$

Proof. Let
$$U = \mathbf{i}, \qquad V = \mathbf{j}, \qquad q = 1.$$
Then
$$VUq = \mathbf{ji} = -\mathbf{k},$$
whereas
$$UVq = \mathbf{ij} = \mathbf{k}.$$
Thus
$$VUq \neq UVq.$$
$\square$

10.1 Political-economic interpretation

The theorem formalizes path dependence. Examples include:

$$\text{monetary easing} \to \text{capital controls} \neq \text{capital controls} \to \text{monetary easing},$$

$$\text{sanctions} \to \text{supply-chain rerouting} \neq \text{supply-chain rerouting} \to \text{sanctions},$$
$$\text{military mobilization} \to \text{diplomacy} \neq \text{diplomacy} \to \text{military mobilization}.$$

11 Macroeconomics

11.1 Quaternionic aggregate demand and supply

Let aggregate demand and supply be

$$\mathcal{D}_t = D_{0,t} + \mathbf{i}D_{A,t} + \mathbf{j}D_{B,t} + \mathbf{k}D_{AB,t},$$

$$\mathcal{S}_t = S_{0,t} + \mathbf{i}S_{A,t} + \mathbf{j}S_{B,t} + \mathbf{k}S_{AB,t}.$$

The macroeconomic gap is
$$\mathcal{G}_t = |\mathcal{D}_t - \mathcal{S}_t|.$$
Stability requires both magnitude compatibility and orientation compatibility:

$$|\mathcal{D}_t| \approx |\mathcal{S}_t|, \qquad C(\mathcal{D}_t, \mathcal{S}_t) \approx 1.$$

11.2 Two-economy output system

For economies A and B,

$$\log Y_{A,t} = a_A + \alpha_A \log|\mathcal{K}_{A,t}| + \beta_A \log|\mathcal{L}_{A,t}| + \tau_A \log|\mathcal{T}_{AB,t}| - \lambda_A D_{A,t} + \varepsilon_{A,t},$$

$$\log Y_{B,t} = a_B + \alpha_B \log|\mathcal{K}_{B,t}| + \beta_B \log|\mathcal{L}_{B,t}| + \tau_B \log|\mathcal{T}_{BA,t}| - \lambda_B D_{B,t} + \varepsilon_{B,t},$$

where
$$D_{A,t} = D(\mathcal{K}_{A,t}, \mathcal{L}_{A,t}) + D(\mathcal{M}_{A,t}, \mathcal{P}_{A,t}) + D(\mathcal{P}_{A,t}, \mathcal{P}_{B,t}),$$
and
$$D_{B,t} = D(\mathcal{K}_{B,t}, \mathcal{L}_{B,t}) + D(\mathcal{M}_{B,t}, \mathcal{P}_{B,t}) + D(\mathcal{P}_{B,t}, \mathcal{P}_{A,t}).$$

12 Money and Finance

12.1 Monetary quaternion

Define the monetary-financial quaternion of economy A:

$$\mathcal{M}_A = M_A^r + \mathbf{i}E_A + \mathbf{j}X_{A\leftarrow B} + \mathbf{k}S_{AB}.$$

Here

$$M_A^r = \text{measured liquidity},$$

$$E_A = \text{domestic expectation and credibility},$$

$$X_{A\leftarrow B} = \text{foreign financial exposure},$$

$$S_{AB} = \text{bilateral stress: sanctions, swaps, debt, exchange pressure.}$$

A phase-aware inflation equation is

$$\pi_{A,t+1} = \beta_0 + \beta_1 \Delta \log |\mathcal{M}_{A,t}| + \beta_2 D(\mathcal{M}_{A,t}, \mathcal{P}_{A,t}) + \beta_3 D(\mathcal{M}_{A,t}, \mathcal{M}_{B,t}) + \beta_4 x_{A,t} + \varepsilon_{A,t+1}.$$

The term

$$D(\mathcal{M}_{A,t}, \mathcal{P}_{A,t})$$

measures domestic monetary-policy incoherence. The term

$$D(\mathcal{M}_{A,t}, \mathcal{M}_{B,t})$$

measures bilateral monetary-financial misalignment.

12.2 Bank runs and creditor alignment

Let creditor n have withdrawal-orientation quaternion $u_{n,t} \in S^3$. Define the creditor order parameter

$$R_t = \left| \frac{1}{N} \sum_{n=1}^{N} u_{n,t} \right|.$$

If creditors align on withdrawal, then R_t rises. A bank-run condition can be written as

$$R_t > R_c \quad \text{and} \quad \frac{\text{liquid reserves}_t}{\text{demand liabilities}_t} < \rho_c.$$

Thus a bank run is a balance-sheet event plus a quaternionic coordination event.

13 Exchange Rates

Let $e_{AB,t}$ be units of currency A per unit of currency B. A quaternionic exchange-rate model is

$$\Delta \log e_{AB,t+1} = a_0 + a_1(\pi_{A,t} - \pi_{B,t}) + a_2(r_{A,t} - r_{B,t}) - a_3 C(\mathcal{M}_{A,t}, \mathcal{M}_{B,t}) + a_4 D(\mathcal{P}_{A,t}, \mathcal{P}_{B,t}) + \varepsilon_{t+1}.$$

Monetary coherence stabilizes the exchange rate:

$$C(\mathcal{M}_{A,t}, \mathcal{M}_{B,t}) \uparrow \quad \Rightarrow \quad |\Delta \log e_{AB,t+1}| \downarrow,$$

while institutional decoherence destabilizes it:

$$D(\mathcal{P}_{A,t}, \mathcal{P}_{B,t}) \uparrow \quad \Rightarrow \quad |\Delta \log e_{AB,t+1}| \uparrow.$$

14 Trade and Supply Chains

Let $\mathcal{S}_{A,t}$ be economy A's supply field and $\mathcal{D}_{B,t}$ economy B's demand field. Bilateral trade from A to B is

$$T_{A\to B,t} = T_0|\mathcal{S}_{A,t}|^{\mu}|\mathcal{D}_{B,t}|^{\nu}\exp\left[-\chi D(\mathcal{S}_{A,t},\mathcal{D}_{B,t}) - \zeta D(\mathcal{M}_{A,t},\mathcal{M}_{B,t}) - \xi D(\mathcal{P}_{A,t},\mathcal{P}_{B,t})\right].$$

Trade is maximized when supply, demand, monetary, and institutional orientations are coherent.

Proposition 4 (Trade coherence)**.** *For fixed amplitudes, $T_{A\to B,t}$ is maximized when*

$$C(\mathcal{S}_{A,t},\mathcal{D}_{B,t}) = C(\mathcal{M}_{A,t},\mathcal{M}_{B,t}) = C(\mathcal{P}_{A,t},\mathcal{P}_{B,t}) = 1.$$

Proof. The proof is identical to the coherence-production theorem. Each decoherence term is nonnegative and enters the exponential negatively. Hence trade is maximized when each decoherence term is zero. □

15 Political Science and Institutions

Let the political-institutional field of economy A be

$$\mathcal{P}_A = P_A^r + \mathbf{i}L_A + \mathbf{j}R_{A\leftarrow B} + \mathbf{k}H_{AB},$$

where P_A^r is administrative state capacity, L_A is domestic legitimacy, $R_{A\leftarrow B}$ is reputational exposure to economy B, and H_{AB} is bilateral diplomatic orientation.

Regime stability may be modeled as

$$S_{A,t} = \sigma\left[\gamma_0 + \gamma_1|\mathcal{P}_{A,t}| - \gamma_2 D(\mathcal{P}_{A,t},\mathcal{M}_{A,t}) - \gamma_3 D(\mathcal{P}_{A,t},\mathcal{G}_{A,t})\right],$$

where

$$\sigma(x) = \frac{1}{1+e^{-x}}.$$

A legitimacy crisis occurs when

$$|\mathcal{P}_{A,t}| < P_c$$

or

$$D(\mathcal{P}_{A,t},\mathcal{L}_{A,t}) > D_c,$$

where $\mathcal{L}_{A,t}$ is the citizen legitimacy field.

16 Geopolitics and International Relations

Let geopolitical power be

$$\mathcal{G}_A = G_A^r + \mathbf{i}I_A + \mathbf{j}R_{A\leftarrow B} + \mathbf{k}H_{AB}.$$

Here G_A^r is material geopolitical power, I_A is internal ideological coherence, $R_{A\leftarrow B}$ is relative orientation toward economy B, and H_{AB} is alliance or rivalry intensity.

16.1 Coalition power

For a coalition $\mathcal{C}$, define

$$\mathcal{G}_{\mathcal{C}} = \sum_{a \in \mathcal{C}} \mathcal{G}_a.$$

Then

$$|\mathcal{G}_{\mathcal{C}}|^2 = \sum_{a \in \mathcal{C}} |\mathcal{G}_a|^2 + 2 \sum_{a<b} |\mathcal{G}_a||\mathcal{G}_b| C(\mathcal{G}_a, \mathcal{G}_b).$$

Theorem 4 (Quaternionic coalition theorem)**.** *Coalition power is not additive in member capacity. It is amplified by positive coherence and diminished by negative coherence.*

Proof. The result follows by repeated application of the bilateral interference theorem to

$$\mathcal{G}_{\mathcal{C}} = \sum_{a \in \mathcal{C}} \mathcal{G}_a.$$

Each cross term contributes

$$2|\mathcal{G}_a||\mathcal{G}_b| C(\mathcal{G}_a, \mathcal{G}_b).$$

Positive coherence raises coalition power; negative coherence lowers it. □

16.2 Conflict risk

For economies A and B, define deterrence balance:

$$\mathcal{D}_{AB} = \log \left(\frac{|\mathcal{G}_A|}{|\mathcal{G}_B|} \right) + \delta C(\mathcal{G}_A, \mathcal{G}_{A,\text{allies}}) - \eta D(\mathcal{P}_A, \mathcal{P}_B).$$

Conflict probability is

$$\Pr(\text{conflict}_{AB}) = \frac{1}{1 + \exp[-(\gamma_0 - \gamma_1 \mathcal{D}_{AB})]}.$$

17 Warfare Economics

Let military-strategic capacity be

$$\mathcal{W}_A = W_A^r + \mathbf{i} D_A + \mathbf{j} F_{A \leftarrow B} + \mathbf{k} C_{AB}.$$

Here W_A^r denotes hardware, troops, and logistics; D_A denotes doctrine, morale, and command quality; $F_{A \leftarrow B}$ denotes foreign pressure or adversary exposure; and C_{AB} denotes cyber, informational, and strategic interaction.

17.1 Quaternionic contest success

The probability that A defeats B is

$$\Pr(A \text{ defeats } B) = \frac{1}{1 + \exp[-Z_{AB}]},$$

where

$$Z_{AB} = \beta_0 + \beta_1 \log \left(\frac{|\mathcal{W}_A|}{|\mathcal{W}_B|} \right) + \beta_2 C(\mathcal{W}_A, \mathcal{P}_A) - \beta_3 C(\mathcal{W}_B, \mathcal{P}_B) + \beta_4 C(\mathcal{G}_A, \mathcal{G}_{A,\text{allies}}) - \beta_5 D(\mathcal{W}_A, \mathcal{G}_A).$$

Military success therefore depends on relative amplitude, domestic political-military coherence, adversary coherence, alliance coherence, and alignment between battlefield means and geopolitical ends.

17.2 Information warfare as orientation attack

Information warfare modifies orientation rather than merely reducing amplitude:

$$u_{\mathcal{W}_A,t+1} = R_{B\to A,t}u_{\mathcal{W}_A,t},$$

where $R_{B\to A,t}$ is an adversarial quaternionic rotation. Its purpose is to increase

$$D(\mathcal{W}_A, \mathcal{P}_A)$$

or

$$D(\mathcal{W}_A, \mathcal{G}_A).$$

18 Sanctions and Economic Warfare

Let sanctions imposed by B on A be represented by a quaternionic operator

$$\mathcal{S}_{B\to A} = s_0 + \mathbf{i}s_A + \mathbf{j}s_B + \mathbf{k}s_{AB}.$$

The post-sanction transmission field is

$$\mathcal{T}_{AB,t+1} = \mathcal{S}_{B\to A}\mathcal{T}_{AB,t}.$$

Sanctions can reduce amplitude:

$$|\mathcal{T}_{AB,t+1}| < |\mathcal{T}_{AB,t}|,$$

rotate orientation:

$$C(\mathcal{T}_{AB,t+1}, \mathcal{T}_{AB,t}) < 1,$$

or do both.

A sanctions effectiveness equation is

$$E_{B\to A,t} = \delta_0 + \delta_1 \log\left(\frac{|\mathcal{S}_{B\to A,t}|}{|\mathcal{R}_{A,t}|}\right) + \delta_2 D(\mathcal{T}_{AB,t}, \mathcal{M}_{A,t}) + \delta_3 C(\mathcal{G}_{B,t}, \mathcal{G}_{B,\text{allies},t}) + \varepsilon_t,$$

where $\mathcal{R}_{A,t}$ is economy A's resilience field.

19 Statistical Mechanics of Quaternionic Economies

Let N agents or institutions have unit quaternion orientations

$$u_{1,t}, \dots, u_{N,t} \in S^3.$$

Define the order parameter

$$R_t = \left|\frac{1}{N}\sum_{n=1}^{N} u_{n,t}\right|.$$

Then

$$0 \le R_t \le 1.$$

Proposition 5 (Order parameter bounds)**.** *For unit quaternions $u_{n,t}$,*

$$0 \le R_t \le 1.$$

Proof. By the triangle inequality,

$$\left|\frac{1}{N}\sum_{n=1}^{N} u_{n,t}\right| \le \frac{1}{N}\sum_{n=1}^{N} |u_{n,t}| = 1.$$

Nonnegativity follows from the definition of norm. □

19.1 Interaction energy

Define the quaternionic coordination energy

$$U_t = -\frac{J}{2N} \sum_{m,n=1}^{N} C(u_{m,t}, u_{n,t}),$$

where $J > 0$ measures coordination strength.

Proposition 6 (Coherence minimizes energy)**.** *The energy U_t is minimized when all orientations are equal.*

Proof. Since

$$C(u_m, u_n) \leq 1,$$

the sum

$$\sum_{m,n} C(u_m, u_n)$$

is maximized when

$$C(u_m, u_n) = 1$$

for all m, n, which occurs when all orientations are equal. Since U_t is the negative of this sum multiplied by a positive constant, U_t is minimized at full coherence. □

19.2 Crisis as quaternionic decoherence

Define the system as crisis-prone when

$$R_t < R_c.$$

A bilateral systemic crisis occurs when

$$R_{AB,t} < R_c \quad \text{and} \quad |\mathcal{T}_{AB,t}| < T_c.$$

Thus crisis requires both low coherence and weak transmission capacity.

20 Econometrics and Identification

The econometrician observes real data:

$$X_t = \mathcal{O}(\Psi_t) + \varepsilon_t,$$

but the full quaternionic state $\Psi_t \in \mathbb{H}^n$ is latent.

20.1 Proxy construction

A quaternionic field can be estimated by constructing proxies:

$$\widehat{\mathcal{X}}_t = \widehat{X}_{0,t} + \mathbf{i}\widehat{X}_{A,t} + \mathbf{j}\widehat{X}_{B,t} + \mathbf{k}\widehat{X}_{AB,t}.$$

Coordinate	Meaning	Possible proxies
X_0	common material baseline	GDP, capital stock, liquidity, trade volume
X_A	A-latent component	surveys, institutional quality, patents, skills
X_B	B-latent component	surveys, institutional quality, patents, skills
X_{AB}	bilateral field	trade exposure, FDI, sanctions, alliance scores

Table 3: Proxy construction for quaternionic fields.

20.2 Panel estimation

For a panel of country pairs (A, B), estimate

$$\log Y_{A,t} = a_A + \alpha_A \log |\mathcal{K}_{A,t}| + \beta_A \log |\mathcal{L}_{A,t}| + \tau_A \log |\mathcal{T}_{AB,t}|$$
$$-\lambda_{KL,A} D(\mathcal{K}_{A,t}, \mathcal{L}_{A,t}) - \lambda_{MP,A} D(\mathcal{M}_{A,t}, \mathcal{P}_{A,t}) - \lambda_{AB,A} D(\mathcal{P}_{A,t}, \mathcal{P}_{B,t}) + \mu_A + \nu_t + \varepsilon_{A,t}.$$

Here μ_A is an economy fixed effect and ν_t is a time effect.

20.3 Testable hypotheses

Hypothesis	Empirical restriction
Domestic factor coherence raises output	$\lambda_{KL,A} > 0$
Monetary-political incoherence raises inflation	$\beta_2 > 0$
Bilateral monetary coherence stabilizes exchange rates	$a_3 > 0$
Institutional decoherence reduces trade	$\xi > 0$
Alliance coherence raises deterrence	$\delta > 0$
Military-political incoherence lowers victory probability	$\beta_5 > 0$

Table 4: Testable implications of Quaternion Economic Field Theory.

21 Algorithms

21.1 Algorithm 1: Quaternionic field construction

Input: Data for X_0, X_A, X_B, X_{AB}.
Step 1: Normalize each coordinate to comparable scale.
Step 2: Construct

$$\mathcal{X} = X_0 + \mathbf{i}X_A + \mathbf{j}X_B + \mathbf{k}X_{AB}.$$

Step 3: Compute amplitude

$$|\mathcal{X}| = \sqrt{X_0^2 + X_A^2 + X_B^2 + X_{AB}^2}.$$

Step 4: Compute orientation

$$u_{\mathcal{X}} = \frac{\mathcal{X}}{|\mathcal{X}|}.$$

Output: Quaternionic field, capacity, and orientation.

`The following space was deliberately left blank.`

21.2 Algorithm 2: Quaternionic production estimation

Input: $Y_A, \mathcal{K}_A, \mathcal{L}_A, \mathcal{M}_A, \mathcal{P}_A, \mathcal{P}_B, \mathcal{T}_{AB}$.
Step 1: Compute norms of all quaternionic fields.
Step 2: Compute decoherence terms:

$$D(\mathcal{K}_A, \mathcal{L}_A), \quad D(\mathcal{M}_A, \mathcal{P}_A), \quad D(\mathcal{P}_A, \mathcal{P}_B).$$

Step 3: Estimate

$$\log Y_A = a_A + \alpha_A \log |\mathcal{K}_A| + \beta_A \log |\mathcal{L}_A| + \tau_A \log |\mathcal{T}_{AB}| - \sum_m \lambda_m D_m + \varepsilon_A.$$

Step 4: Test whether $\lambda_m > 0$.
Output: Amplitude elasticities and coherence penalties.

21.3 Algorithm 3: Bilateral crisis detection

Input: Unit orientations $u_{1,t}, \ldots, u_{N,t}$, bilateral transmission field $\mathcal{T}_{AB,t}$, thresholds R_c, T_c.
Step 1: Compute

$$R_t = \left| \frac{1}{N} \sum_{n=1}^{N} u_{n,t} \right|.$$

Step 2: Compute transmission amplitude $|\mathcal{T}_{AB,t}|$.
Step 3: If $R_t < R_c$ and $|\mathcal{T}_{AB,t}| < T_c$, classify the pair as crisis-prone.
Output: Coherence index, transmission index, and crisis flag.

22 AI and Machine Learning

22.1 Quaternionic prediction

Let the target field be $q_t \in \mathbb{H}$ and the forecast be $\widehat{q}_t \in \mathbb{H}$. A quaternion-aware loss is

$$\mathcal{L} = \sum_t \left[(|\widehat{q}_t| - |q_t|)^2 + \lambda_D D(\widehat{q}_t, q_t) \right] + \rho \|\Theta\|^2.$$

This loss penalizes both capacity error and orientation error.

22.2 Quaternion neural layer

Let $h_\ell \in \mathbb{H}^n$. A quaternionic neural layer is

$$h_{\ell+1} = \sigma(W_\ell h_\ell + b_\ell),$$

where

$$W_\ell \in \mathbb{H}^{m \times n}, \qquad b_\ell \in \mathbb{H}^m.$$

The multiplication is Hamilton multiplication. A componentwise activation may be written as

$$\sigma(q) = \sigma_0(q_0) + \mathbf{i}\sigma_1(q_1) + \mathbf{j}\sigma_2(q_2) + \mathbf{k}\sigma_3(q_3).$$

22.3 Quaternion reinforcement learning

Let the state be

$$\Psi_t \in \mathbb{H}^n,$$

and let policies $u_{A,t}, u_{B,t} \in \mathbb{R}^m$ affect the transition:

$$\Psi_{t+1} = F_\Theta(\Psi_t, u_{A,t}, u_{B,t}) + \eta_t.$$

A planner maximizes

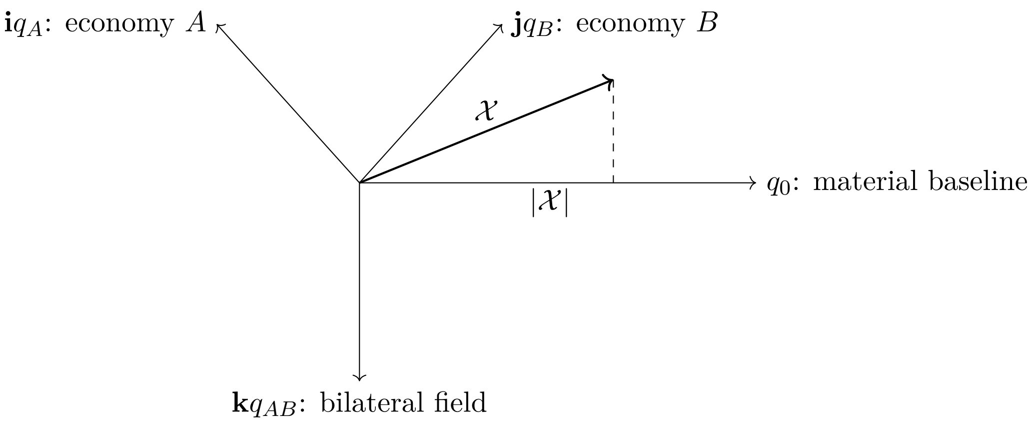
$$\mathbb{E} \sum_{s=t}^{\infty} \delta^{s-t} \mathcal{U}_s,$$

where

$$\mathcal{U}_t = w_Y(\log Y_{A,t} + \log Y_{B,t}) - w_\pi(\pi_{A,t}^2 + \pi_{B,t}^2) - w_D D_{AB,t} - w_C C_t.$$

23 Vector Graphics

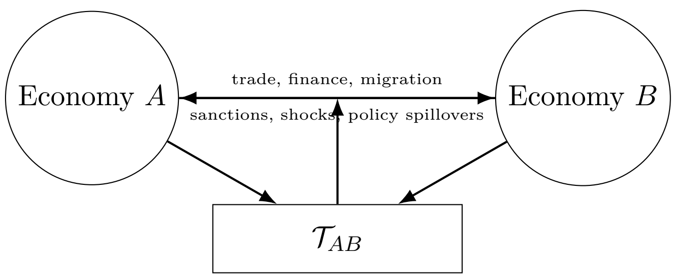
23.1 Quaternionic two-economy field



Quaternionic field: baseline plus two domestic latent fields plus bilateral interaction.

Figure 1: A quaternionic economic field.

23.2 The two-economy transmission field



The bilateral field is not reducible to either domestic economy alone.

Figure 2: The bilateral transmission field in a two-economy quaternionic model.

23.3 Non-commutative policy sequence

$$\boxed{q_t} \xrightarrow{U} \boxed{Uq_t} \xrightarrow{V} \boxed{V(Uq_t)} \qquad VUq_t$$

$$\boxed{q_t} \xrightarrow{V} \boxed{Vq_t} \xrightarrow{U} \boxed{U(Vq_t)} \qquad UVq_t$$

$VUq_t \neq UVq_t$ in general: policy sequence matters.

Figure 3: Non-commutative policy operations.

24 Limitations

Quaternion Economic Field Theory has several limitations.

First, the four coordinates of a quaternionic field are not directly observed. They require proxy construction, factor extraction, survey data, institutional data, market data, or machine-learning inference.

Second, quaternionic multiplication is non-commutative. This is economically useful but mathematically demanding. Modelers must specify whether transformations act on the left, on the right, or on both sides.

Third, coherence is not always normatively desirable. High coherence can produce growth, institutional capacity, and alliance strength, but it can also produce bubbles, mass mobilization, authoritarian consolidation, or destructive war.

Fourth, the bilateral component q_{AB} must be carefully identified. Without empirical discipline, it may become a residual category for unexplained bilateral effects.

Fifth, policy models must distinguish between amplitude effects and orientation effects. A policy may increase capacity while lowering coherence, or increase coherence while reducing capacity.

25 Conclusion

Quaternion Economic Field Theory generalizes complex economic modeling from one economy to two interacting economies. The central object is

$$q = q_0 + \mathbf{i}q_A + \mathbf{j}q_B + \mathbf{k}q_{AB}.$$

The q_0-coordinate represents material baseline capacity. The $\mathbf{i}q_A$-coordinate represents economy A's latent internal field. The $\mathbf{j}q_B$-coordinate represents economy B's latent internal field. The $\mathbf{k}q_{AB}$-coordinate represents the bilateral field: trade, finance, sanctions, supply chains, migration, alliances, rivalry, and information conflict.

The framework's main conceptual claim is:

$$\boxed{\text{A two-economy system is not } A + B; \text{ it is } A + B + AB.}$$

The main mathematical claim is that bilateral capacity depends on both amplitude and quaternionic coherence:

$$|Q_A + Q_B|^2 = |Q_A|^2 + |Q_B|^2 + 2|Q_A||Q_B|C(Q_A, Q_B).$$

The main policy claim is that order matters:

$$UV \neq VU.$$

Thus the sequencing of monetary policy, sanctions, industrial policy, diplomacy, war, and financial intervention may alter the final state of the two-economy system.

Quaternion Economic Field Theory provides a unified language for two-economy macroeconomics, finance, exchange rates, trade, geopolitics, warfare economics, statistical mechanics, econometrics, and AI/ML. It extends the idea of economic phase from a circle to a three-sphere, and it elevates bilateral interaction from an external linkage to a structural field.

References

[1] S. Ghosh. *Complex Economic Field Theory*. Manuscript, 2026.

[2] W. R. Hamilton. On quaternions; or on a new system of imaginaries in algebra. *Philosophical Magazine*, 25:489–495, 1844.

[3] J. B. Kuipers. *Quaternions and Rotation Sequences*. Princeton University Press, 1999.

[4] K. J. Arrow and G. Debreu. Existence of an equilibrium for a competitive economy. *Econometrica*, 22(3):265–290, 1954.

[5] J. M. Keynes. *The General Theory of Employment, Interest and Money*. Macmillan, 1936.

[6] J. F. Muth. Rational expectations and the theory of price movements. *Econometrica*, 29(3):315–335, 1961.

[7] R. E. Lucas. Expectations and the neutrality of money. *Journal of Economic Theory*, 4(2):103–124, 1972.

[8] F. A. Hayek. The use of knowledge in society. *American Economic Review*, 35(4):519–530, 1945.

[9] D. C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, 1990.

[10] D. Acemoglu and J. A. Robinson. *Why Nations Fail*. Crown, 2012.

[11] T. C. Schelling. *The Strategy of Conflict*. Harvard University Press, 1960.

[12] C. von Clausewitz. *On War*. Princeton University Press edition, 1976.

[13] F. W. Lanchester. *Aircraft in Warfare: The Dawn of the Fourth Arm*. Constable, 1916.

[14] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.

[15] Y. Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.

[16] E. Ising. Beitrag zur Theorie des Ferromagnetismus. *Zeitschrift fur Physik*, 31:253–258, 1925.

[17] D. W. Diamond and P. H. Dybvig. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419, 1983.

[18] P. Krugman. A model of balance-of-payments crises. *Journal of Money, Credit and Banking*, 11(3):311–325, 1979.

[19] J. D. Hamilton. *Time Series Analysis.* Princeton University Press, 1994.

[20] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning.* MIT Press, 2016.

[21] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction.* MIT Press, 2018.

[22] A. Hirose. *Complex-Valued Neural Networks.* Springer, 2012.

[23] T. Parcollet, M. Morchid, and G. Linares. Quaternion convolutional neural networks for heterogeneous image processing. *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2019.

Glossary

Amplitude
The norm $|q|$ of a quaternionic economic field, interpreted as effective capacity.

Bilateral field
The $\mathbf{k}q_{AB}$ component of a quaternionic field, representing trade, finance, sanctions, alliances, rivalry, migration, supply chains, cyber interaction, or other cross-economy channels.

Coherence
The normalized scalar product

$$C(p, q) = \frac{\mathrm{Re}(p\overline{q})}{|p||q|},$$

measuring alignment between two quaternionic fields.

Decoherence
The distance

$$D(p, q) = 1 - C(p, q),$$

measuring misalignment between two quaternionic fields.

Hamilton product
The non-commutative multiplication rule of quaternions.

Material baseline
The real coordinate q_0 of a quaternionic field.

Non-commutative policy
A policy environment in which the order of operations matters:

$$UV \neq VU.$$

Quaternion
An element

$$q = q_0 + \mathbf{i}q_1 + \mathbf{j}q_2 + \mathbf{k}q_3$$

of the algebra $\mathbb{H}$.

Quaternionic orientation
The unit quaternion

$$u_q = q/|q|,$$

representing direction in material-latent-bilateral space.

Realization map
A function

$$\mathcal{R} : \mathbb{H}^n \to \mathbb{R}^m$$

that converts quaternionic field states into observed real-valued outcomes.

Transmission field
The quaternionic field $\mathcal{T}_{AB}$ representing bilateral economic, financial, technological, political, and strategic linkages.

The End